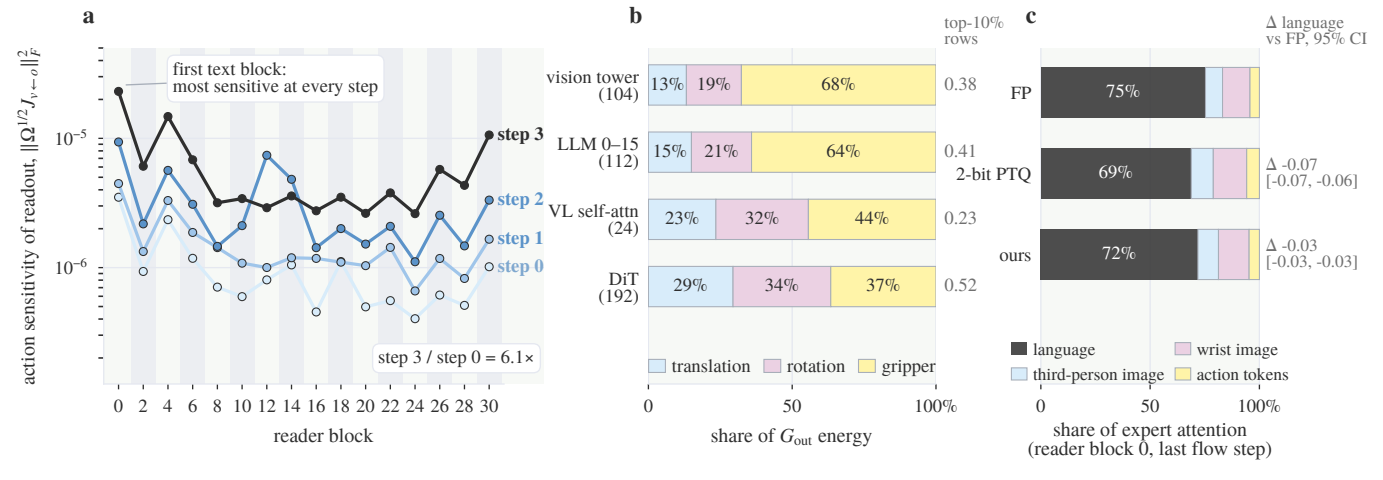
a
b
c
action sensitivity of readout, $\|\Omega^{1/2} J_{v\leftarrow o}\|_F^2$
first text block:
most sensitive at every step
step 3
step 2
step 1
step 0
step 3 / step 0 = 6.1×
10^{-5}
10^{-6}
0 2 4 6 8 10 12 14 16 18 20 22 24 26 28 30
reader block
top-10% rows
vision tower (104)
LLM 0–15 (112)
VL self-attn (24)
DiT (192)
13%
19%
68%
0.38
15%
21%
64%
0.41
23%
32%
44%
0.23
29%
34%
37%
0.52
translation
rotation
gripper
0
50
100%
share of G_{out} energy
Δ language vs FP, 95% CI
FP
75%
2-bit PTQ
69%
Δ -0.07 [-0.07, -0.06]
ours
72%
Δ -0.03 [-0.03, -0.03]
language
wrist image
third-person image
action tokens
0
50
100%
share of expert attention
(reader block 0, last flow step)